

Alternate method for Rule 1:

↳ For: 1st order ODEs

↳ The DE: $Mdx + Ndy = 0$ can be rearranged to

the form: $\boxed{\frac{dy}{dx} + P(x)y = Q(x)}$

← This is called the standard form of a Linear DE.

* Here $P(x)$ and $Q(x)$ are functions solely in terms of x

↳ For this DE, the integrating factor is:

$$IF = e^{\int P(x) dx}$$

← You can cross-check this using Rule 1.

↳ Multiplying the IF with the standard form of the DE, we get:

$$e^{\int P(x) dx} \cdot \frac{dy}{dx} + P(x) \cdot e^{\int P(x) dx} \cdot y = e^{\int P(x) dx} \cdot Q(x)$$

$$\Rightarrow \int \left(e^{\int P(x) dx} \cdot \frac{dy}{dx} + P(x) \cdot e^{\int P(x) dx} \cdot y \right) dx = \int e^{\int P(x) dx} \cdot Q(x) dx$$

↳ Integrating the LHS automatically gives us:

$$e^{\int P(x) dx} \cdot y$$

← Notice that differentiating this gives us the integrand on the LHS (u.v method)

↳ So the DE becomes:

$$\underbrace{e^{\int P(x) dx}}_{IF} \cdot y = \int e^{\int P(x) dx} \cdot Q(x) dx$$

Now we only need to integrate the RHS to get the solution of the DE.

↳ This is called the variation of parameters method.

(e.g.) Find the general solution of the following DE:

$$x \cdot \frac{dy}{dx} + 4y = x^3 - x$$

Give the largest interval over which the general solution is defined. Determine whether there are any transient terms in the general solution.

Solⁿ Now: $x \cdot \frac{dy}{dx} + 4y = x^3 - x$

$$\Rightarrow \frac{dy}{dx} + \frac{4}{x} \cdot y = x^2 - 1 \quad \leftarrow \text{Standard form}$$

From the above eqⁿ, we can see that:

$$P(x) = \frac{4}{x} \quad \text{and} \quad Q(x) = x^2 - 1$$

$$\therefore \text{IF} = e^{\int \frac{4}{x} dx} = e^{4 \ln x} = x^4$$

$$x^4 \cdot \frac{dy}{dx} + 4x^3 \cdot y = x^6 - x^4$$

$$\Rightarrow \int \left(x^4 \cdot \frac{dy}{dx} + 4x^3 \cdot y \right) dx = \int (x^6 - x^4) dx$$

$$\Rightarrow x^4 \cdot y = \frac{x^7}{7} - \frac{x^5}{5} + C \quad \leftarrow \text{General Solution}$$

↳ Interval over which both $P(x)$ and $Q(x)$ are continuous: $(-\infty, 0)$ and $(0, +\infty)$

↳ Transient terms:

$$x^4 \cdot y = \frac{x^7}{7} - \frac{x^5}{5} + C$$

$$\Rightarrow y = \frac{x^3}{7} - \frac{x}{5} + \frac{C}{x^4}$$

Transient terms are those which disappear (approach zero) as $x \rightarrow +\infty$

From the above eqⁿ, as $x \rightarrow +\infty$, $\frac{C}{x^4} \rightarrow 0$

So $\frac{C}{x^4}$ is a transient term. //

P.T.O.

Bernoulli's DE:

↳ For: 1st order, non-linear ODEs

↳ When the DE is of the form:

$$\frac{dy}{dx} + P(x) \cdot y = Q(x) \cdot y^n$$

A DE of this form can be converted into the standard form of a linear DE using the substitution $u = y^{1-n}$.

$$u = y^{1-n} \Rightarrow \frac{du}{dx} = (1-n) y^{-n} \cdot \frac{dy}{dx}$$

$$\text{Then the DE becomes: } \frac{dy}{dx} + P(x) \cdot y = Q(x) \cdot y^n$$

$$\Rightarrow y^{-n} \cdot \frac{dy}{dx} + P(x) \cdot y^{1-n} = Q(x)$$

$$\Rightarrow \frac{1}{(1-n)} \cdot \frac{du}{dx} + P(x) \cdot u = Q(x)$$

$$\Rightarrow \frac{du}{dx} + (1-n) \cdot P(x) \cdot u = (1-n) \cdot Q(x)$$

Here, the dependent variable here has changed from "y" to "u".
The DE has also changed from non-linear to linear.

(e.g.) Solve the non-linear ODE: $\frac{dy}{dx} + \frac{2}{x} \cdot y = x^2 y^3$

Solⁿ Let $u = y^{1-3} = y^{-2}$

$$\therefore \frac{du}{dx} = -2y^{-3} \frac{dy}{dx} \Rightarrow y^{-3} \cdot \frac{dy}{dx} = -\frac{1}{2} \cdot \frac{du}{dx}$$

$$\hookrightarrow \text{ODE: } \frac{dy}{dx} + \frac{2}{x} \cdot y = x^2 y^3$$

$$\Rightarrow y^{-3} \frac{dy}{dx} + \frac{2}{x} \cdot y^{-2} = x^2$$

$$\Rightarrow -\frac{1}{2} \cdot \frac{du}{dx} + \frac{2}{x} \cdot u = x^2$$

$$\Rightarrow \frac{du}{dx} - \frac{4}{x} \cdot u = -2x^2 \quad \leftarrow \text{The non-linear ODE in } y \text{ has changed into a linear ODE in } u.$$

$$\uparrow \text{Of the form: } \frac{dy}{dx} + P(x) \cdot y = Q(x)$$

$$\text{Let IF} = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x}$$

$$= e^{\ln x^{-4}} = \frac{1}{x^4}$$

$$\therefore \text{Soln to DE: } \frac{1}{x^4} \cdot u = \int \frac{1}{x^4} \cdot -2x^2 dx = \int \frac{-2}{x^2} dx = \frac{2}{x} + C$$

$$\Rightarrow \frac{u}{x^4} = \frac{2}{x} + C \quad \Rightarrow \frac{y^{-2}}{x^4} = \frac{2}{x} + C$$

$$\Rightarrow \frac{1}{x^3 y^2} = 2 + Cx$$

$$\Rightarrow y^2 = \frac{1}{x^3(2+Cx)} \quad \leftarrow C \text{ is the constant of integration.}$$

(e.g.) Solve the ODE: $\frac{dy}{dx} + y = xy^2$. $\leftarrow$ H.W.

Try this one yourselves!

$\rightarrow$ Next, we shall look at ways of using these different DE solving methods, to solve actual, real world problems.

The problems here are limited to linear, 1st order DEs.

Applications of 1st Order ODEs: $\leftarrow$ Bacteria Growth Problem

(e.g.) A culture initially has P_0 number of bacteria.

At $t=1h$, the number of bacteria is measured to be $\frac{3P_0}{2}$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

Soln $\rightarrow$ Number of bacteria present at time, $t = P(t)$ $\leftarrow$ dependent
 $\uparrow$ independent

* Always identify the independent and dependent variables first

$\rightarrow$ Initial values: $P(0) = P_0$ and $P(1) = \frac{3P_0}{2}$

→ Growth rate of bacteria $\propto$ No. of bacteria present

$$\Rightarrow \frac{dP(t)}{dt} \propto P(t)$$

$$\Rightarrow \frac{dP}{dt} = k \cdot P \quad \leftarrow k = \text{Constant of proportionality}$$

This is the DE we need to solve.

→ Solving this DE:

$$\frac{dP}{dt} = kP \quad \Rightarrow \quad \frac{1}{P} dP = k \cdot dt$$

$$\Rightarrow \int \frac{1}{P} dP = \int k \cdot dt \quad \Rightarrow \quad \ln P = kt + \ln C$$

$$\Rightarrow \ln P = \ln e^{kt} + \ln C \quad \Rightarrow \quad \ln P = \ln (C \cdot e^{kt})$$

$$\Rightarrow P = C \cdot e^{kt}$$

→ Initially, at time $t=0$, $P(0) = P_0$.

$$\therefore P_0 = C \cdot e^{k \cdot 0}$$

$$\Rightarrow C = P_0$$

Hence, the equation becomes:

$$P(t) = P_0 \cdot e^{kt}$$

→ At time $t=1$, $P(1) = \frac{3}{2} P_0$. So from the above eqⁿ:

$$\therefore \frac{3}{2} \cancel{P_0} = \cancel{P_0} \cdot e^{k \cdot 1}$$

$$\Rightarrow \frac{3}{2} = e^k$$

$$\Rightarrow \ln \frac{3}{2} = k$$

Hence, $P(t) = P_0 \cdot e^{\ln \frac{3}{2} \cdot t}$ ← The solution to our DE.

→ To find the time at which $P(t) = 3P_0$:

$$3P_0 = P_0 \cdot e^{\ln \frac{3}{2} \cdot t}$$

$$\Rightarrow 3 = e^{\ln \frac{3}{2} \cdot t}$$

$$\Rightarrow \ln 3 = \ln\left(\frac{3}{2}\right) \cdot t \quad \Rightarrow t = \frac{\ln 3}{\ln(\frac{3}{2})}$$

$$\Rightarrow t = 2.71 \text{ (to 2 d.p.)}$$

Radioactive Decay

(e.g.) A breeder reactor converts relatively stable Uranium-238 into the isotope Plutonium-239. After 15 years, it's determined that 0.043% of the initial amount A_0 of Plutonium has disintegrated. Find the half-life of the isotope if the rate of disintegration is proportional to the amount remaining.

Soln → Amount of Plutonium present at time, $t = A(t)$ ← dependent

→ Initial values: $A(0) = A_0$

$$\text{and } A(15) = A_0 - 0.00043 A_0$$

$$= 0.99957 A_0$$

$$0.043\% \text{ of } A_0$$

$$= \frac{0.043}{100} \times A_0$$

$$= 0.00043 A_0$$

→ Disintegration rate $\propto$ Amount of Plutonium remaining

$$\Rightarrow -\frac{dA(t)}{dt} \propto A(t) \quad \leftarrow \text{LHS is negative here since}$$

amount of Plutonium is decreasing

$$\Rightarrow -\frac{dA}{dt} = k \cdot A$$

This is the DE we need to solve.

$$\Rightarrow \int \frac{1}{A} dA = \int -k dt \quad \Rightarrow \ln A = -kt + \ln C$$

$$\Rightarrow \ln A = \ln C \cdot e^{-kt} \quad \Rightarrow A = C \cdot e^{-kt}$$

$$\rightarrow \text{Given } A(0) = A_0,$$

$$A_0 = C \cdot e^{-k \cdot 0} \quad \Rightarrow C = A_0$$

$$\therefore A = A_0 e^{-kt}$$

$$\rightarrow \text{Now, } A(15) = 0.99957 A_0$$

$$\Rightarrow 0.99957 A_0 = A_0 e^{-15k}$$

$$\Rightarrow e^{-15k} = 0.99957$$

$$\Rightarrow k = -\frac{\ln(0.99957)}{15}$$

$$\text{Hence, } A = A_0 e^{\frac{\ln(0.99957)}{15} t}$$

$$\text{To find half-life, let } A = \frac{A_0}{2}. \therefore \frac{A_0}{2} = A_0 e^{\frac{\ln(0.99957)}{15} t}$$

$$\Rightarrow t = \frac{15 \ln(1/2)}{\ln(0.99957)} \approx 24174.35383 //$$

Newton's Law of Cooling / Warming:

Rate at which the temperature of an object changes $\propto$ Difference between the temperature of the object and the temperature of the surroundings

$$\Rightarrow \frac{dT}{dt} \propto (T - T_m)$$

$$\Rightarrow \frac{dT}{dt} = k(T - T_m) \quad \leftarrow \text{Here, } k < 0.$$

$\hookrightarrow T = \text{Temperature of the object}$

$T_m = \text{Temperature of the surrounding medium.}$

(e.g.) When a cake is removed from an oven, its temperature is measured to be 300°F . Then 3 minutes later, its temperature becomes 200°F . Find the temperature function $T(t)$ when the room temperature is 70°F .

Solⁿ $\rightarrow$ Temperature of the cake at time, $t = T(t)$
 $\rightarrow$ Initial values: $T(0) = 300$ and $T(3) = 200$
 $\rightarrow$ From Newton's Law of Cooling, we know:

$$\frac{dT}{dt} \propto (T - T_m) \quad \text{where } T_m = 70$$

$$\Rightarrow \frac{dT}{dt} = k(T - 70)$$

This is the DE we need to solve.

$$\therefore \frac{1}{T-70} dT = k dt$$

$$\Rightarrow \ln(T-70) = kt + \ln C$$

$$\Rightarrow \ln(T-70) = \ln e^{kt} + \ln C$$

$$\Rightarrow T-70 = C \cdot e^{kt} \quad \Rightarrow T = C \cdot e^{kt} + 70$$

$$\text{Now, } T(0) = 300$$

$$\Rightarrow C \cdot e^0 + 70 = 300$$

$$\Rightarrow C = 230$$

$$\text{Also, } T(3) = 200$$

$$\Rightarrow 230e^{3k} + 70 = 200 \quad \Rightarrow e^{3k} = \frac{130}{230}$$

$$\Rightarrow 3k = \ln\left(\frac{13}{23}\right) \Rightarrow k = \frac{1}{3} \ln\left(\frac{13}{23}\right)$$

$$\therefore T = 230 e^{\frac{1}{3} \ln\left(\frac{13}{23}\right)t} + 70$$

Application of DEs in solving Circuit Problems:

First order DEs can be used to model two kinds of electrical circuits:

i) LR Circuits

ii) RC Circuits.

But before we proceed, we need to review one of Kirchhoff's Laws of electrical circuits.

→ Kirchhoff's 2nd Law (Voltage Law):

The total sum of all electric potentials or voltages around any closed circuit loop is zero.

i) LR circuits:

An LR-circuit or Inductor-Resistor circuit is one which has both an inductor and a resistor.

* Inductor: → Represented by the circuit symbol .

→ It is an electrical component that resists the change in current in a circuit.

↳ if current tries to rise, it pushes back to lower it.

↳ if current tries to drop, it releases stored charge to increase it.

→ There is either a rise or drop in electric potential across an inductor based on whether the current tries to drop or rise, respectively. So we can state:

$$V \propto \frac{dI}{dt} \quad \text{across an inductor}$$

* V = voltage
I = current

$$\Rightarrow V = L \cdot \frac{dI}{dt}$$

← The proportionality constant L is the inductance of the inductor.

→ The unit of measurement of inductance is Henry (H).

(e.g.) A 30V emf is applied to an LR series circuit in which the inductance is 0.1H and the resistance is 50Ω . Find the current $I(t)$ if the current was initially 0A. Determine the current as $t \rightarrow \infty$.

Solⁿ: Potential rises: 30V from the power supply
Potential drops: Across the resistance, $V = IR$
 $= I \times 50 = 50I$

$$\begin{aligned}\text{Across the inductor, } V &= L \times \frac{dI}{dt} \\ &= 0.1 \times \frac{dI}{dt} \\ &= \frac{1}{10} \cdot \frac{dI}{dt}\end{aligned}$$

Applying Kirchhoff's 2nd Law:

$$+30 - 50I - \frac{1}{10} \cdot \frac{dI}{dt} = 0$$

$$\Rightarrow \frac{1}{10} \cdot \frac{dI}{dt} + 50I = 30$$

$$\Rightarrow \frac{dI}{dt} + 500I = 300 \quad \leftarrow \text{DE of the form } \frac{dI}{dt} + P(t) \cdot I = Q(t)$$

$$\therefore IF = e^{\int 500 dt} = e^{500t}$$

So, solution to the DE:

$$IF \cdot I(t) = \int IF \cdot Q(t) dt$$

$$\begin{aligned}\Rightarrow e^{500t} \cdot I &= \int e^{500t} \cdot 300 dt \\ &= \frac{e^{500t}}{500} \cdot 300 + C \\ &= \frac{3}{5} e^{500t} + C\end{aligned}$$

$$\text{At } t=0, I=0 \longrightarrow 0 = \frac{3}{5} \cdot e^0 + C \Rightarrow C = -\frac{3}{5}$$

$$\therefore \text{Solⁿ: } e^{500t} \cdot I = \frac{3}{5} e^{500t} - \frac{3}{5}$$

$$\Rightarrow I = \left(\frac{3}{5} e^{500t} - \frac{3}{5} \right) \cdot e^{-500t}$$

$$\Rightarrow I(t) = \frac{3}{5} - \frac{3}{5} \cdot e^{-500t}$$

$$\begin{aligned} \rightarrow \text{As } t \rightarrow \infty: \quad & \lim_{t \rightarrow \infty} I(t) \\ &= \lim_{t \rightarrow \infty} \left(\frac{3}{5} - \frac{3}{5} \cdot e^{-500t} \right) \\ &= \frac{3}{5} \text{ A} \end{aligned}$$

* Capacitor: $\rightarrow$ Represented by the circuit symbol $\text{---} \text{||} \text{---}$.

$\rightarrow$ An electrical component that stores charge.

$\hookrightarrow$ When it starts charging, it initially has a potential of 0V. Due to the potential difference with the power supply, current flows and it starts to accumulate charge. As it charges up, its potential difference with the battery decreases, causing current flow to decrease, till it eventually becomes zero.

$\hookrightarrow$ When it starts discharging, there initially is high flow of current due to the high potential difference across the capacitor, which acts as emf for the current in the circuit. But as the capacitor is losing charge, the potential difference across it decreases, and the emf driving current around the discharging circuit decreases to eventually becoming zero. Hence, the current gradually decreases to zero here as well.

$\rightarrow$ The charge holding capacity of a capacitor is given by its capacitance.

$\rightarrow$ Potential drop across or emf provided by a capacitor (for charging and discharging circuits respectively) is given by:

* Q = charge stored by capacitor (in coulombs (C))

$$V = \frac{Q}{C}$$

$\leftarrow C$ is the capacitance of the capacitor.

$\rightarrow$ The unit of measurement of the capacitance of a capacitor is Farads (F).

e.g. A 100V emf is applied to an RC series circuit in which the resistance is 200Ω and the capacitance is 10^{-4}F . Find the charge $Q(t)$ on the capacitor if initially there is 0 charge stored. Also find the current $I(t)$.

Solⁿ: Since emf is being applied, this is a charging circuit.

Potential rises: 100V from the power supply.

Potential drops: Across the resistance: $V = IR = I \times 200 = 200I$

Across the capacitor: $V = \frac{Q}{C} = \frac{Q}{10^{-4}} = 10000Q$

Since $Q = It \Rightarrow I = \frac{Q}{t}$, current here depends on the charge stored by the capacitor. So Q is the variable we need to work with.

Also, since Q is changing with time, we can express I as the rate of change in Q with respect to "t" as: $I = \frac{dQ}{dt}$.

Applying Kirchhoff's 2nd Law: $+100 - 200I - 10000Q = 0$

$$\Rightarrow 100 = 200 \cdot \frac{dQ}{dt} + 10000Q$$

$$\Rightarrow \frac{100}{200} = \frac{10000}{200} \times Q + \frac{dQ}{dt}$$

$$\Rightarrow \frac{dQ}{dt} + 50Q = \frac{1}{2} \quad \leftarrow \text{DE of the form: } \frac{dy}{dx} + P(x) \cdot y = Q(x)$$

$$\therefore \text{IF} = e^{\int 50 dt} = e^{50t}$$

So solⁿ of the DE:

$$\begin{aligned} e^{50t} \cdot Q(t) &= \int e^{50t} \cdot \frac{1}{2} dt \\ &= \frac{1}{2} \times \frac{1}{50} \times e^{50t} + C \\ &= \frac{1}{100} \cdot e^{50t} + C \end{aligned}$$

At $t=0$, $Q=0 \rightarrow$ DE becomes:

$$0 = \frac{1}{100} \times 1 + C \Rightarrow C = -\frac{1}{100}$$

$$\therefore e^{50t} \cdot Q(t) = \frac{1}{100} e^{50t} - \frac{1}{100}$$

$$\Rightarrow Q(t) = \frac{1}{100} - \frac{1}{100} \cdot e^{-50t}$$

$$\rightarrow \text{We know, } Q = It \Rightarrow I = \frac{Q}{t} = \frac{\left(\frac{1}{100} - \frac{1}{100} \cdot e^{-50t} \right)}{t}$$

$$\Rightarrow I(t) = \frac{1}{100t} - \frac{1}{100t} \cdot e^{-50t} //$$